GRE Quantitative Reasoning

모의고사 Set 10 — 문제지

최신 문제(뒷번호) · S2 고난도

구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내
- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 10 해설지)에 있습니다.

답안 기록표

S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
답															

Section 1 12 Questions · 21 Minutes

Question 1 대소비교

522. Polygon P has n sides (n>4).
Quantity A: 180 more than the sum of the degree measures of the interior angles of P
Quantity B: The sum of the degree measures of the interior angles of a polygon with n+1 sides

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

1039. $|2x+7|<13$
Quantity A: x^2
Quantity B: 9

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

583. n is the least possible integer such that 108n is the square of an integer.
Quantity A: n
Quantity B: 9

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

1144. A survey of voters who were either supporters of Candidate A or supporters of Candidate B showed that 64 percent of the supporters of Candidate A and 50 percent of the supporters of Candidate B were in favor of a progressive income tax.
Quantity A: The percent of the voters surveyed who were in favor of a progressive
Quantity B: 57%

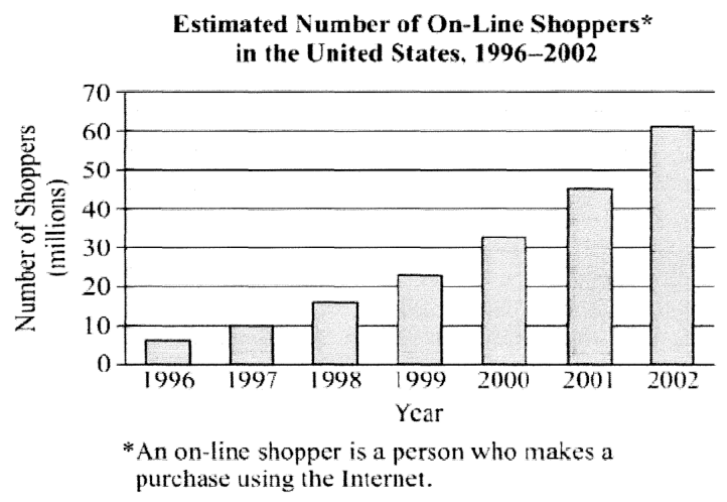
(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 객관식

856. In the rectangular coordinate system, a certain line has slope 3. Which of the following pairs of points could be on the line?
A (-1, 2) and (-2, 5)
B(0, 1) and (3, 2)
C(0, 1) and (3, 4)
D(3, 1) and (4, 4)
E(3, 1) and (5, 6)

(A) (B) (C) (D) (E)

Questions 6 to 8 are based on the following data.



Question 6 객관식

970. The estimated number of on-line shoppers for 1996 is approximately what percent of the estimated number for 2001?
A2%
B13%
C22%
D34%
E41%

(A) (B) (C) (D) (E)

Question 7 객관식

971. For 1998, if the total amount of online sales was 6.0 billion dollars, which of the following is closest to the average estimated amount spent per online shopper? (1 billion = 1,000 million)
A$265
B$300
C$375
D$450
E$600

A B C D E

Question 8 객관식

972. From 2001 to 2002 the percent increase in the estimated number of on-line shoppers is approximately which of the following?
A15%
B25%
C35%
D45%
E55%

A B C D E

Question 9 객관식

985. If the function f is defined by $f(x)=1-2^x$ for all integers x, then $-f(10)$ is closest to which of the following?
A-1,000
B-100
C0
D100
E1,000

A B C D E

Question 10 주관식

✓ 1103. An order for bottles of vitamins from a certain mail order company costs $12.04 per bottle plus a shipping cost of $4.80 regardless of the number of bottles ordered. Over the past year, the company has received 100 orders for bottles of vitamins. The standard deviation of the numbers of bottles per order for the 100 orders is 1.5 bottles. What is the standard deviation of the 100 costs for the orders?

$ [ ]

Question 11 복수선택

751. The degree measure of each angle of a regular polygon with n sides is between 100 and 130. Which of the following could be the value of n?
Indicate all such integers.
A4
B5
C6
D7
E8
F9
G10

Indicate all that apply. A B C D E F G

Question 12 객관식

1130. x is a positive integer
Which of the following x could ensure that 10^x is closest to 2^{20}?
A4
B6
C8
D10
E12

A B C D E

Section 2 15 Questions · 26 Minutes

Question 1 대소비교

936. Points S and T are lies on the segment PQ, the distance of PQ is 16, the distance of PS is 7, the distance of ST is 2.
Quantity A: The distance of PT
Quantity B: The distance of QT

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

734. $3x+2y=8$
x and y are integers and $x > y$.
Quantity A: $3x+4y$
Quantity B: 8

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

602. k is randomly selected from the set of integers from 1 to 100, inclusive.
Quantity A: The probability that $(k)(k+1)(k+2)(k+3)(k+4)$ is divisible by 20
Quantity B: 1

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

585. The positive numbers a, b, c, d, e, and f are different from each other.
Quantity A: The range of the numbers a, b, c, d, e, and f
Quantity B: The range of the numbers a^2, b^2, c^2, d^2, e^2 and f^2

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 대소비교

339. A set of k consecutive integers, including 2. The sum of the integers in the set is -11.
Quantity A: k
Quantity B: 10

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 6 객관식

1030. A rectangular solid P has height 2c and a base of width a and length b. Two other rectangular solids, Q and R, each have height c and bases of width a and length b. Which of the following represents the amount by which the sum of the surface areas of Q and R exceeds the surface areas of P?
A2ab
B4ab
C2ab+2bc
D2ab+4ac
E2ab+4ac+4bc

(A) (B) (C) (D) (E)

Question 7 객관식

671. For which of the following values of x is the units digit of the product $(2)(3^x)$ equal to 4?
A12
B13
C14
D15
E16

(A) (B) (C) (D) (E)

Question 8 객관식

1263. The weekly sales commission at Company X is 2 percent of the first $2,000 of weekly sales plus 15 percent of the weekly sales in excess of the first $2,000. The weekly sales commission at Company Y is 5 percent of the first $2,000 of weekly sales plus 10 percent of the weekly sales in excess of the first $2,000. In a given week, what is the amount of weekly sales at Company Y that would earn the same weekly sales commission as $5,000 in weekly sales at Company X?
A.$5,900
B.$5,300
C.$4,900
D.$4,800
E.$4,500

(A) (B) (C) (D) (E)

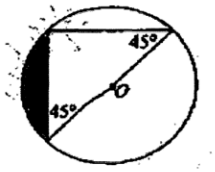
Question 9 복수선택

676. If p is an odd integer, and if 5 is a factor of $p+p^2$,which of the following might be the remainder when p is divided by 5? Indicate all such numbers.
A0
B1
C2
D3
E4

Indicate all that apply. A B C D E

Question 10 주관식

1276.



A triangle is inscribed in the circle with center O as shown above. If the radius of the circle is 5, what is the area of the shaded region? Give your answer to the nearest integer.

Question 11 객관식

368. If k and n are each positive integers between 12 and 30, then (5+k)/(7+n) will be equal to 5/7 for how many pairs of (k, n)?
ATwo
BThree
CFour
DSeventeen
ENineteen

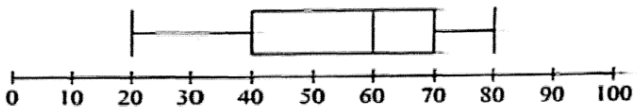
A B C D E

Question 12 객관식

786. A pound of tea makes 210 cups of tea, and a pound of coffee makes 40 cups of coffee. If, on the average, a restaurant serves 12 times as many cups of coffee as tea, what is the ratio of pounds of tea to pounds of coffee used at the restaurant?
A1/63
B6/125
C1/12
D4/21
E16/7

A B C D E

Question 13 복수선택



357. The boxplot above summarizes a list of 240 numbers. Which of the following statements must be true? Indicate all such expressions.
AThe list contains the number 60.
BThe range of the numbers in the list is less than 4 times the intequartile range of the numbers.
CThere were more numbers in the list that are less than 40 than there are numbers that are greater than 70.

Indicate all that apply. A B C

Question 14 객관식

766. Set A={9, 13, 4, 2, 7}
Set B={9, 13, 4, 2, 9}
Set C={9, 13, 4, 2, 4}
Set D={9, 13, 4, 2, 6}
Which of the following statement illustrates the exact order of standard deviation of these sets of numbers from the lowest to the greatest?
AA < B < C < D
BA < B < D < C
CD < B < C < A
DC < B < D < A
EA < D < B < C

A B C D E

Question 15 객관식

593. A group of n people is categorized as follows.
9 people are taller than 6 feet.
14 people are shorter than 5(1/2) feet.
12 people are under 21 years old.
If each person in the group is in at least one of the three categories, then n can be any integer between
A9 and 14, inclusive
B12 and 21, inclusive
C14 and 23, inclusive
D21 and 26, inclusive
E23 and 35, inclusive

A B C D E
